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References:

- "Industrial Guide to Ergonomics Engineering", Dr. Osama Mohammed Elmardi Suleiman Khayal
- Mechanical Engineering Department, Faculty of Engineering and Technology, Nile Valley University,
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Faculty HOD Dean of Academics 21/1/26

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Lecture-1

Factors affecting performance

Fitness and health

The passage emphasizes the duty of care professionals have in assessing their own fitness to perform their duties. This is legally required for roles like doctors, flight crew, air traffic controllers, and now licensed aircraft maintenance engineers. Since some physical or mental health issues may not be immediately noticeable, teamwork plays a crucial role in identifying substandard performance or unusual behavior. Colleagues and supervisors should report concerns to management so appropriate support and counseling can be provided, ensuring safety and efficiency in the workplace.

Maintaining fitness and health in daily life is essential for overall well-being. A balanced lifestyle that includes physical activity, proper nutrition, mental well-being, and adequate rest can improve quality of life and longevity (Positive Measures)

Positive Measures

Take common sense steps to maintain their fitness and health. These include:

• Eating regular meals and a well-balanced diet;

• Taking regular exercise (exercise sufficient to double the resting pulse rate for 20 minutes, three times a week is often recommended);

• Stopping smoking;

• Sensible alcohol intake (for men, this is no more than 3 - 4 units a day or 28 per week,

where a unit is equivalent to half a pint of beer or a glass of wine or spirit)

Many conditions can impact on the health and fitness of an engineer and there is not space here to offer a complete list. However, such a list would include:

• Minor physical illness (such as colds, flu, etc.);

• More major physical illness (such as malaria, etc.);

• Mental illness (such as depression, etc.);

• Minor injury (such as a sprained wrist, etc.);

• Major injury (such as a broken arm, etc.);

Stress: domestic and work related

Stress can be defined as any force, that when applied to a system, causes some significant modification of its form, where forces can be physical, psychological or due to social pressures.

A. Domestic Stress Causes:

- Family conflicts or relationship issues

- Financial problems
- Health concerns (personal or family)
- Household responsibilities and parenting pressures
- Loss of a loved one
- Lack of personal time and relaxation

B. Work-Related Stress Causes:

- Heavy workload and tight deadlines
- Job insecurity or fear of job loss
- Conflicts with colleagues or management
- Lack of work-life balance
- High-pressure work environment
- Unclear job expectations

2. Symptoms/ Effects of Stress

A. Physical Symptoms:

- Headaches and muscle tension
- Fatigue and sleep disturbances
- High blood pressure and heart palpitations
- Weakened immune system
- Stomach issues (nausea, indigestion)

B. Emotional and Mental Symptoms:

- Anxiety and irritability
- Depression and mood swings
- Difficulty concentrating
- Feelings of overwhelm or helplessness
- Loss of motivation

C. Behavioural Symptoms:

- Withdrawal from social interactions
- Increased use of alcohol, caffeine, or smoking
- Poor work performance or absenteeism
- Changes in eating habits (overeating or loss of appetite)

Stress Management

Involves two main strategies: defense (managing symptoms) and coping (addressing the root cause). Coping strategies focus on resolving stressors, such as delegating tasks or

prioritizing work. However, in situations beyond personal control (e.g., emergencies), individuals can use proven techniques to manage stress, including:

1. Relaxation methods (meditation, deep breathing)
2. Balanced sleep and diet
3. Regular physical activity
4. Counseling (talking to friends, colleagues, or professionals)

There is no instant cure for stress, but practical steps and a common-sense approach can help manage it effectively.

Time pressure and deadlines

Time pressure refers to the stress and urgency felt when tasks must be completed within a limited time. Deadlines are specific time limits set for completing work, which can create both motivation and stress.

Causes of Time Pressure and Deadline Stress:

- Heavy workload with tight schedules
- Poor time management or procrastination
- Unexpected changes or last-minute tasks
- Unrealistic expectations from managers or clients
- Lack of resources or team support

Effects:

- Increased stress and anxiety
- Reduced concentration and quality of work
- Physical health issues (fatigue, headaches)
- Lower job satisfaction and burnout

Reference

- https://www.slideshare.net/slideshow/human-factors-topic-1-introduction/18379460#11
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Lecture-2

Managing Time Pressure and Deadlines:

- Prioritize tasks based on urgency and importance
- Break work into smaller steps to manage progress
- Set realistic goals and avoid overcommitting
- Use time management techniques like the Pomodoro Technique or to-do lists
- Delegate tasks when possible
- Stay organized with schedules and reminders

Workload - overload and underload

Workload refers to the amount of tasks or responsibilities assigned to an individual. Both overload (too much work) and underload (too little work) can negatively impact performance and well-being.

1. Work Overload

Occurs when the workload exceeds an individual's capacity, leading to stress and burnout.

Causes:

- Excessive responsibilities or unrealistic deadlines
- Staff shortages or inefficient task distribution
- Poor time management or lack of delegation
- High expectations from supervisors or clients

Effects:

- Increased stress, fatigue, and anxiety
- Reduced work quality and productivity
- Physical health issues (headaches, high blood pressure)
- Burnout and job dissatisfaction

Management:

- Prioritize and delegate tasks effectively
- Use time management strategies
- Take regular breaks to maintain efficiency
- Communicate workload concerns with supervisors

2. Work Underload

Happens when there is too little work, leading to boredom and decreased motivation.

Causes:

- Job roles with minimal responsibilities
- Lack of skill utilization
- Poor task distribution in a team
- Organizational inefficiencies

Effects:

- Decreased motivation and engagement
- Feelings of frustration or lack of purpose
- Lower productivity and career stagnation
- Increased absenteeism or job dissatisfaction

Management:

- Seek additional tasks or responsibilities
- Learn new skills or take on professional development
- Engage in creative or side projects
- Communicate with management about workload concerns

Arousal

Arousal refers to a state of alertness and activation in the body and mind, influencing performance, motivation, and emotions. It is regulated by the nervous system and can range from low (drowsiness) to high (intense excitement or stress).

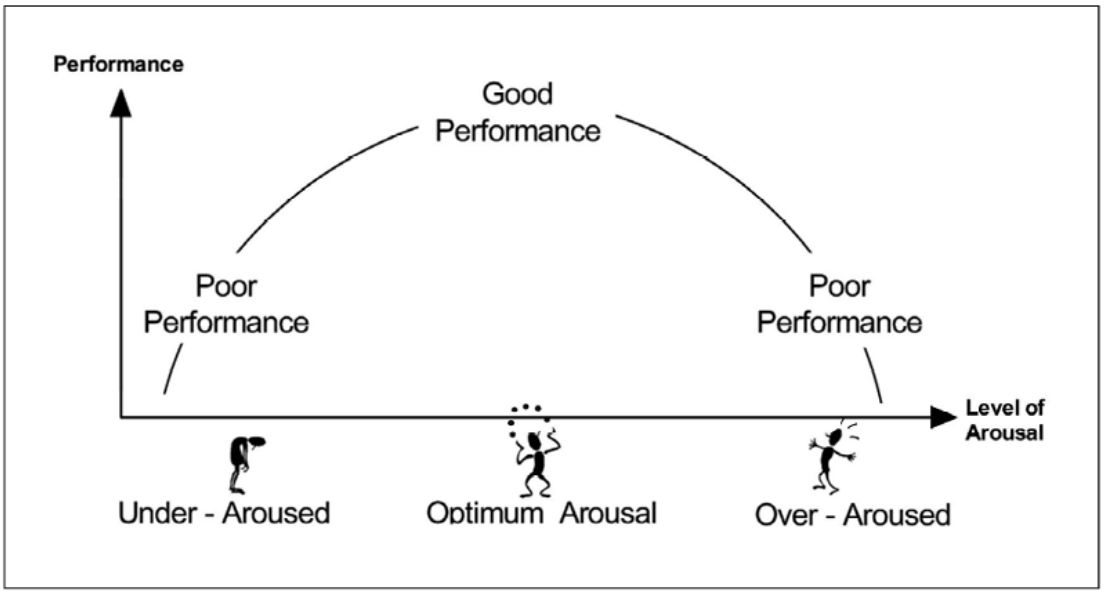


Fig-1 Optimum arousal leads to best task performance (adapted from Thom 1999)

- The x-axis represents the **level of arousal** (from low to high).
- The y-axis represents **performance** (poor to good).
- The curve follows an **inverted U-shape**, showing that performance improves with arousal up to a point, but declines if arousal becomes too high.

Stages of Arousal:

1. **Under-Aroused (Low Arousal, Poor Performance)**
    - Lack of motivation, boredom, and low energy.
    - Performance suffers due to inattention and lack of effort.
2. **Optimum Arousal (Moderate Arousal, Best Performance)**
    - Ideal balance of focus, energy, and motivation.
    - Enhances concentration, productivity, and efficiency.
3. **Over-Aroused (High Arousal, Poor Performance)**
    - Excessive stress, anxiety, or nervousness.
    - Performance declines due to panic, errors, and impaired decision-making

Types of Arousal:

1. Physiological Arousal – Involves physical changes like increased heart rate, sweating, and muscle tension (e.g., during exercise or stress).

2. Cognitive Arousal – Involves mental alertness and focus (e.g., when solving problems or learning new skills).
3. Emotional Arousal – Involves feelings like excitement, fear, or anxiety.

Effects of Arousal on Performance:

- Optimal Arousal: Moderate levels enhance focus and efficiency (Yerkes-Dodson Law).
- Low Arousal: Leads to boredom, low motivation, and poor performance.
- High Arousal: Can cause anxiety, stress, and reduced efficiency.

Managing Arousal:

- Increase Arousal: Engage in physical activity, listen to upbeat music, or set challenging goals.
- Decrease Arousal: Practice deep breathing, meditation, or relaxation techniques.

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Lecture-3

Factors Determining Workload

Workload is influenced by various factors that affect an individual's capacity to complete tasks efficiently. These factors can be categorized into individual, organizational, and external influences.

1. Individual Factors

- Skill and Experience: More skilled employees handle tasks more efficiently.
- Physical and Mental Capacity: Fatigue, stress, or health issues affect workload capacity.
- Time Management Skills: Poor time management can lead to inefficiency.

2. Organizational Factors

- Job Role and Responsibilities: Clearly defined roles help distribute workload effectively.
- Task Complexity: More complex tasks require greater effort and time.
- Resource Availability: Lack of tools, technology, or staff increases workload pressure.
- Work Environment: A supportive and efficient workplace reduces workload stress.

3. External Factors

- Industry Demands: Some industries (e.g., healthcare, aviation) naturally have high workloads.
- Economic Conditions: Budget cuts or staff shortages increase workload.
- Regulatory Requirements: Compliance and legal obligations may add to workload.

Sleep

Sleep is a natural state of rest in which the body and mind go through essential recovery processes. It is characterized by reduced consciousness, decreased sensory activity, and inactivity of voluntary muscles. Sleep is vital for maintaining overall health, cognitive function, and physical well-being.

A good rule of thumb is that one hour of high-quality sleep is good for two hours of activity.

Stages of Sleep:

1. **Non-Rapid Eye Movement (NREM) Sleep:**
    - Stage 1: Light sleep, transition from wakefulness.
    - Stage 2: Heart rate and body temperature drop.
    - Stage 3: Deep sleep, essential for physical recovery.
2. **Rapid Eye Movement (REM) Sleep:**
    - Dreaming occurs.
    - Brain activity increases, important for memory and learning.

Functions of Sleep:

- Restores energy and repairs body tissues.
- Enhances memory, learning, and cognitive function.
- Strengthens the immune system.
- Regulates mood and emotional well-being.

Effects of Sleep Deprivation:

- Reduced concentration and poor decision-making.
- Increased stress, anxiety, and irritability.
- Higher risk of heart disease, obesity, and weakened immunity.

Recommended Sleep Duration:

- Adults: 7–9 hours
- Teenagers: 8–10 hours
- Children: 9–12 hours

Fatigue

Fatigue is a state of physical, mental, or emotional exhaustion that reduces a person's ability to perform tasks efficiently. It can result from prolonged work, lack of sleep, stress, or health conditions.

Types of Fatigue:

1. **Physical Fatigue:**

- Muscle weakness and tiredness due to excessive physical activity.
- Reduced stamina and slower reflexes.

2. **Mental Fatigue:**
    - Difficulty concentrating and making decisions.
    - Memory problems and lack of motivation.
3. **Emotional Fatigue:**
    - Feeling overwhelmed, anxious, or irritable.
    - Burnout from prolonged stress or emotional strain.

Causes of Fatigue:

- **Lack of Sleep:** Poor sleep quality or insufficient rest.
- **Workload and Stress:** Long hours, high pressure, or repetitive tasks.
- **Poor Nutrition and Dehydration:** Lack of essential nutrients and fluids.
- **Medical Conditions:** Anemia, diabetes, heart disease, or depression.
- **Lifestyle Factors:** Sedentary behavior, excessive alcohol or caffeine consumption.

Effects of Fatigue:

- Reduced work performance and productivity.
- Increased risk of mistakes and accidents.
- Weak immune system and long-term health issues.
- Mood swings, anxiety, and depression.

Managing Fatigue:

- Maintain a consistent sleep routine (7–9 hours for adults).
- Take short breaks during work to avoid mental exhaustion.
- Stay hydrated and eat a balanced diet.
- Engage in regular physical activity to boost energy.
- Manage stress through relaxation techniques like meditation or deep breathing.

Reference

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Shift work

Shift work refers to a work schedule that falls outside the traditional 9-to-5 hours. It includes evening, night, rotating, or irregular shifts, commonly used in industries that require 24-hour operations, such as healthcare, transportation, manufacturing, and emergency services.

Types of Shift Work:

1. **Fixed Shifts:** Employees work the same shift consistently (e.g., permanent night shifts).
2. **Rotating Shifts:** Employees alternate between different shifts (e.g., morning, evening, and night).
3. **Split Shifts:** Work is divided into two periods within the same day.
4. **On-Call Shifts:** Employees are available to work when needed, often with unpredictable hours.

Challenges of Shift Work:

- **Disrupts the body's natural circadian rhythm**, leading to sleep disturbances.
- **Increases fatigue and mental stress**, affecting productivity and alertness.
- **Higher risk of health issues**, including cardiovascular disease, obesity, and digestive problems.
- **Affects social and family life**, making it difficult to maintain relationships.

Managing Shift Work Effectively:

- Follow a **consistent sleep schedule**, even on off-days.
- Create a **dark, quiet sleep environment** to improve rest quality.
- Maintain a **healthy diet and stay hydrated** to boost energy levels.
- Engage in **regular physical activity** to improve overall well-being.
- Use **strategic napping** before night shifts to reduce fatigue.

Alcohol, medication and drug abuse

Notes by- Karishma Mittal (AP. Civil Engg Department, MITRC, Alwar)

Substance abuse, including alcohol, medication, and illegal drugs, can severely impact physical health, mental well-being, and workplace performance.

Alcohol Abuse

Alcohol abuse refers to excessive or irresponsible drinking that affects daily life and work performance.

Effects of Alcohol Abuse:

- **Short-term:** Impaired judgment, reduced coordination, slowed reaction time, and increased risk of accidents.
- **Long-term:** Liver disease, heart problems, memory loss, mental health disorders, and addiction.
- **Workplace Impact:** Reduced productivity, absenteeism, increased accident risks, and poor decision-making.

Prevention & Management:

- Drink in moderation or avoid alcohol in safety-sensitive jobs.
- Seek professional help for alcohol dependence.
- Follow workplace alcohol policies.

2. Medication Misuse

Medication abuse occurs when prescription or over-the-counter drugs are taken inappropriately (e.g., without a prescription, in excessive doses, or for non-medical reasons).

Commonly Abused Medications:

- **Painkillers (Opioids):** Can lead to addiction, drowsiness, and overdose risks.
- **Sedatives & Sleeping Pills:** Cause drowsiness, confusion, and dependency.
- **Stimulants (e.g., ADHD medications):** Misuse can cause heart issues, anxiety, and insomnia.

Prevention & Management:

- Take medications only as prescribed by a doctor.
- Avoid mixing medications with alcohol or other drugs.
- Seek medical advice before stopping or changing doses.

3. Drug Abuse

Drug abuse involves the misuse of illegal or prescription drugs for recreational purposes, leading to addiction and severe health issues.

Effects of Drug Abuse:

- **Physical:** Heart disease, liver failure, weakened immune system.
- **Mental:** Anxiety, depression, paranoia, hallucinations.
- **Workplace Impact:** Poor performance, safety risks, absenteeism, and legal consequences.

Prevention & Management:

- Education and awareness about the dangers of drug use.
- Access to counseling and rehabilitation programs.
- Strict workplace drug policies and regular drug testing.

Reference

- https://www.slideshare.net/slideshow/human-factors-topic-1-introduction/18379460#11
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